Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/20/2016
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Project Title:	Solution Convection and the Nucleation Precursors in Protein Crystallization		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	10/01/2013	**End Date:**	01/20/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 1/20/2019 per NSSC information (Ed., 1/15/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Conrad, Jacinta Ph.D. (University of Houston) Maes, Dominique Ph.D. (Vrije Universiteit Brussel)		
Grant/Contract No.:	NNX14AD68G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Protein crystallization experiments in microgravity have often generated unexpectedly low or high numbers of crystals; both outcomes may doom experiments designed to obtain a small number of well-separated crystals for x-ray structure studies. This unpredictability has been associated with the suppression of buoyancy-driven convection, as supported by data indicating that protein crystal nucleation rates reach a maximum at an optimal convection velocity. To elucidate the effects of solution convection on crystal nucleation--a crucial issue in several fields of science and engineering--we propose to test two mechanisms: 1. Convection enhances or suppresses the formation of the dense liquid clusters, within which crystal nuclei form. 2. Convection enhances or suppresses the formation of crystal nuclei within the clusters. Tests of mechanism 1. require characterization of the dense liquid clusters at different rates of convection, including its complete absence, only possible in microgravity. We propose to use the Light Microscopy Module (LMM) aboard the International Space Station (ISS) and two variants of the Advanced Colloid Experiment (ACE): ACE-M and ACE-C. To prepare for microgravity experiments, we will design microscopy techniques that can then be implemented on the LMM. We will extend a newly-developed method, differential dynamic microscopy, to quantify the dynamics of the dense liquid clusters. Its variation with solution composition and the rate of convection will be used to identify the cluster formation mechanisms. Towards tests of mechanism 2, we will determine the supersaturation dependencies of the crystal nucleation rates in both quiescent and flowing solutions. For this, we will implement novel microfluidics devices made of SU-8 and glass, which allow long-duration experiments under steady conditions. Finally, we will use the insight on convection effects on nucleation to establish procedures leading to higher perfection of protein crystals and test them with proteins of interest to biomedical sciences.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Solution crystallization underlies a broad range of industrial, laboratory, and physiological processes. Single solution-grown crystals of inorganic salts or mixed organic-inorganic materials are used in non-linear optics elements and for other electronic and optical-electronic applications; chemical products and production intermediates are precipitated as crystals in thousands-of-tons amounts. Another area which relies on solution-grown crystals is pharmacy: the slow crystal dissolution rate is used to achieve sustained release of medications: small-molecules organic, or proteins such as insulin, interferon-alpha, or the human growth hormone. If the administered dose consists of a few equidimensional crystallites, steady medication release rates can be maintained for longer periods than for doses comprised of many smaller crystallites. The formation of protein crystals and crystal-like ordered aggregates underlies several human pathological conditions. An example is the crystallization of hemoglobin C and the polymerization of hemoglobin S that cause, respectively, the CC and sickle cell diseases. The formation of crystals in the eye lens underlies the pathology of several forms of cataract. A unique example of benign protein crystallization in humans and other mammals is the formation of rhombohedral crystals of insulin in the islets of Langerhans in the pancreas. Traditionally, protein crystals have been used for the determination of the atomic structure of protein molecules by x-ray crystallography; this method contributes ~87% of all protein structures solved, with the majority of the other determinations carried out by Nuclear Magnetic Resonance (NMR) spectroscopy. Due to its crucial place at the start of the crystallization process, the nucleation of crystals determines many properties of the emerging crystalline phase. It is obvious that the nucleation selects the polymorphic form and if a different polymorph is desired, conditions at which its nucleation is faster than that of the other possible polymorphs should be sought. If nucleation is fast, many crystals form nearly simultaneously. Their growth depletes the medium of solute and may lead to cessation of nucleation at the later stages of crystallization. Thus, the majority of crystals grow to approximately identical sizes. In contrast, if nucleation is slow and fewer crystals nucleate at a time, the supersaturation in the solution drops slowly, the nucleation of new crystals continues and a population of crystals of various sizes forms. Ultimately, if nucleation is hindered everywhere in the growth container but at a few selected spots, crystals only nucleate at these spots and grow large before the solution is depleted of nutrient. Hence, control of nucleation is a means to control size, size distribution, polymorphism and other properties of the crystals.
	During the reported period, the Vekilov laboratory worked on four research tasks: 1. Develop a method to characterize the protein rich clusters in protein solutions compatible with the light microscopy module (LMM) aboard ISS. 2. Test the role of solution shear for the formation of protein rich clusters. 3. Test the reversibility of cluster formation and study the dynamics of their responses to variation in the external parameters. 4. Test the liquid nature of the clusters. 1. To employ a light microscope for the characterization of the protein rich clusters, we have modified and improved a recent method to study used to quantify dynamics of sub-micrometer particles in solutions from fluctuations in intensity in optical micrographs, differential dynamic microscopy (DDM). Biological objects often scatter light weakly and are frequently smaller than the diffraction limit, complicating measurements of their dynamics. Differential dynamic microscopy is well established for monodispersed particles but has not been applied to polydispersed biological nanoparticles. We used DDM to measure dynamics of protein-rich liquid clusters, which are polydispersed nanoscale objects, in protein solutions, whose size ranged from tens to hundreds of nanometers and whose total volume fraction was less than 10(-5). For solutions of two proteins, lysozyme and hemoglobin A, we measured the dynamics of clusters using DDM and evaluated their diffusion coefficients from the dependence of the diffusion lag time on the scattering wave vector. More recently, we tested the effects of the solution layer thickness on the sensitivity and accuracy of the DDM characterization of cluster dynamics. We found that the optimal thickness is about 200 microm, at which increasing noise compensates stronger b-DDM signals. We tested the DDM method with an illuminator identical to the one to be used aboard ISS. We found that within a broad range of illumination intensities the DDM signal is comparable in signal to noise ratio and accuracy to the one obtained with Kohler illumination in the laboratory. A paper on this improvement of the DDM method has been submitted for publication. 2. We continued our investigations on the effects of shear flow on the properties of the protein rich clusters. We

Task Progress:	employed a gear wheel - motor assembly coupled to a Couette cylindrical cell that allows variation of the solution shear rate from 0.3 s-1 to 200 s-1. Similarly to the findings of the experiments with bubbling induced shear, we found with the protein lysozyme that shear flow increases the average cluster radius and reduces the total volume of the cluster population. We tested the combined effects of shear flow and urea. Urea is a universal denaturant of proteins since it intercalates in the layer of structured water that coats non-polar functional groups along peptide backbone. The resulting weaker hydrophobic attraction between adjacent parts of the peptide chain leads to destabilization of the native protein structure. Depending on the urea concentration, these effects may be manifest as fully denatured peptide chain, or, at lower concentrations, as partially unfolded protein moieties. We employed 1 M urea. Previous experiments in our laboratory, in agreement with published results, have demonstrated that this concentration induces weak destabilization of the native structure of lysozyme and has no effect on the cluster population. However, the application of shear to a lysozyme solution containing 1 M urea led to lower the cluster volume fraction and larger cluster radius at significantly lower shear rates than in the absence of urea. The observed amplification of the shear flow effects in the presence of urea indicates that shear acts by partial protein unfolding, in agreement with the hypothesis formulated in the original proposal. 3. Adjustability of the cluster population characteristics to changing external conditions is a crucial prerequisite for the planned experiments. The solutions to be tested in space will be prepared in the laboratory and, inevitably, the cluster population will reflect the conditions on Earth. To test if the cluster population adjusts to variations in the surrounding conditions, we carried out two types of tests: A) We filtered solutions of lysozyme from different preparations through a filter with 100 nm pore size. We found that this procedure reduces the cluster population in solution of Affimetrix and Sigma lysozyme, but does not affect solutions of Fisher lysozyme. We hypothesize that the reason is the different fluidity of the clusters of different origin: Fisher lysozyme clusters are fluid, while clusters in solutions of Sigma and Affimetrix lysozyme are likely partially gelled. This hypothesis will be tested with the method discussed below. Within 2 days after filtering, the cluster populations in solutions of Affimetrix lysozyme were restored, indicating that their formation is reversible. B) We tested the response of the cluster population to decreasing protein concentration with the protein lysozyme and glucose isomerase. We found that the total volume of the cluster population decreases by 3 orders of magnitude in response to a 10 – 12 x decrease in the solution concentration. The response occurs over 10 min. These observations indicate that population of clusters of these two proteins adjust fast to changing external parameters. 4. To test the liquid nature of the clusters, we developed a novel method to characterize the time scale of shape change of the speckle pattern of a cluster observed in oblique illumination dark field microscope. We crop the movies of cluster Brownian motion to a single cluster. We re-center the images so the center of mass of a cluster is immobile. This insures that all changes of intensity of the cluster speckle pattern are only due to shape variation. With this, the cross-correlation function of the cluster images reveals a characteristic decorrelation time of about 100 ms, consistent for about 10 tested clusters. This time scale is considerable shorter (i.e., by about 100 x) than the characteristic rotation time of a solid cluster of similar size. We conclude that represent the characteristic time of shape change, determined by the cluster surface free energy and viscosity.

Bibliography Type:	Description: (Last Updated: 12/05/2024)
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